Manya Pradeep Narayan

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Education

Cornell University Ithaca, NY

Bachelor of Arts in Computer Science, Minor in Artificial Intelligence *Expected May 2028*

Relevant Coursework: Object Oriented Programming & Data Structures, Functional Programming, Machine Learning, Computer System Organization and Programming, Probability & Statistics, Backend Development (Python, SQL), Discrete Mathematics

Activities: Cornell Culture & Computation Lab (Research Assistant), Cornell Engineers Without Borders (Developer), Women in Computing (Technical Committee), Cornell Data Journal (Analyst), Kappa Theta Pi: Technology Fraternity (VP Marketing)

Experience

Software Engineering Intern: Data, Platform & Product June 2026 – Present

FinMD (Healthcare Lending Intelligence Platform) *Miami, FL / Remote*

- Developed **TERRAIN end-to-end**, integrating housing & physician data into a national risk map for a **$35B bank** demo
- Built a **Python/Scrapy** Doximity crawler enriching a 1.1M-physician registry with education, publication, & career data
- Architected an **LLM pipeline** generating talking points from physician data, with **97.2% grounding** across 250 held-out cases
- Shipped 5 **Next.js** SCOPE modules to prod, integrating **ArcGIS/Postgres** data into payer/competitor insights for bank clients
- Designed an **LLM-powered** interactive market-intelligence feed matching healthcare M&A and policy events to banker portfolios

Research Intern June 2026 – Present

University of Washington Security and Privacy Lab *Seattle, WA*

- Leading an **NSF-funded, first-author** study (paper in progress) on browser-agent susceptibility to deceptive web design
- Built **3 Playwright/CDP harnesses** to autonomously operate browser agents & reconcile traces and server-side ground truth
- Engineered crash-safe **experiment infrastructure** with evidence-integrity checks for unattended multi-machine runs
- Expanded **threat taxonomies** across memory, confirmation gates, temporal manipulation, UI divergence, and recovery behavior
- Identified scaffold failure modes in **Comet, Claude-in-Chrome, etc.** and memory-conditioned risky behavior across **3 models**

Research Assistant April 2026 – Present

University of Washington Human-Computer Interaction Lab *Seattle, WA*

- Built a reusable **LLM pipeline** analyzing Android TalkBack interaction traces for mobile accessibility issues affecting blind users
- Reduced trace-analysis token expenditure **67–81%** by compacting Android view trees into DFS-indexed nodes for issue review
- Found **5 new accessibility failures** across 9 task traces by analyzing navigation through Gmail, Maps, Amazon, and YouTube

Software Engineering Intern September 2025 – April 2026

LessonLoop *Remote*

- Engineered Engagify **(Vue/Nuxt, Node.js, AWS Bedrock)**, an AI lesson revision tool increasing student engagement by 17%
- Implemented a **RAG backend** to retrieve relevant instructional strategies from a 2,500+ item knowledge base for revisions
- Collaborated with a 5-person team to redesign **9+ legacy educator-facing pages**, improving survey and analytics workflows

Projects

Checkpoint | Go, Linux July - August 2026

- Built automatic version control for **AI coding sessions**, preserving transient work and reverting agents independently of Git
- Engineered **Go/Linux daemon** with **fanotify**, process-lineage, and out-of-tree content storage for author-scoped recovery
- Achieved **23 ms** median undo latency and validated recovery across **203** adversarial, integration, and kernel-level tests

Ithaca Reuse App | *Node.js, Express, PostgreSQL* November 2025 – Present

- Developed **Node.js/Express** REST APIs powering donation and sales workflows for Ithaca ReUse NPO (800k+ items annually)
- Built scenario-based Jest tests covering endpoints, PostgreSQL queries, auth, and failure paths, achieving **92% line coverage**

me.json | TypeScript, Node.js, Next.js, React July 2026

- Open-sourced a **npm toolkit** transforming personal websites into AI interfaces via automated extraction and route generation
- Architected a a multi-interface runtime exposing policy-scoped actions through **MCP, REST, TUI, and SDK** surfaces

Symbolic Equation Solver | OCaml April 2026

- Created a symbolic math engine with a **custom LaTeX lexer**, recursive-descent parser, evaluator, and algebraic equation solver
- Added a two-player **TCP math game** with concurrent answer handling, score tracking, and serialized client-server messaging

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, OCaml, SQL, HTML/CSS

Frameworks & Technologies: React/Next.js, Vue/Nuxt, Node.js/Express, Flask, PyTorch, Playwright, Scrapy, Jest

Developer Tools: PostgreSQL, Chrome DevTools Protocol, Docker, AWS, Git